

Location Privacy Challenges and Solutions : Approximate location, a compromise between privacy and usefulness

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Abstract—This research paper examines the anonymity and privacy risks associated with location-based services (LBS). It delves into how approximate location functions in modern devices such as smartphones, and how it can serve as a compromise between user privacy and the practical usefulness of the services. By synthesizing existing strategies and identifying gaps, this study aims to establish a foundation for future research and provide actionable insights into achieving robust privacy protection in LBS ecosystems.

I. INTRODUCTION

The ubiquity of Location-Based Services (LBS), such as Google Maps, has revolutionized how individuals navigate, interact, and access information in real time. While these services offer unparalleled convenience and efficiency, they also introduce significant risks to user anonymity and privacy. The continuous collection, processing, and sharing of location data expose users to potential threats, including unauthorized surveillance, profiling, and malicious exploitation of personal information[1]. These concerns not only decrease user trust but also pose ethical and regulatory challenges for service providers.

This research-based paper will first examine the anonymity and privacy threats inherent in LBS, analyzing the technical and social implications of location data exposure. Additionally, this paper evaluates existing solutions-ranging from user-end strategies like anonymization techniques and privacy settings to intermediary services that aim to mitigate risks without compromising functionality. The goal is to foster a more secure and trustworthy ecosystem for users while not compromising the efficiency of the service.

The structure of this paper is as follows. We will discuss the related work in Section II. In section III we will inspect threats associated with location data and in section IV we will explore user-end solutions, especially Approximate location. Finally in section V we will investigate how the approximate location can be useful in specific case (highly-densated area, countryside, highway) and how it may affect the service.

Problem Definition and Expected Results

The purpose of this study is to be a basis for following research and to help clarify how the different objectives can be accomplished.

Threat Model [2]: The figure 1 states a way to work with LBS, the system architecture here will be the base of our study and we can talk about our threat model. First, we consider a standard GPS service relying on signals from multiple satellites - three for spatial positioning and one for time synchronization - that provides an accurate, unaltered location. In other words, the GPS signal itself is assumed to not be spoofed or manipulated. Second, the user's location is transmitted over common communication networks such as Wi-Fi, 4G, or 5G. We assume that these communication channels are secure and that no adversary is intercepting or monitoring the data in transit. This allows us to exclude network-level attacks from the scope of the analysis.

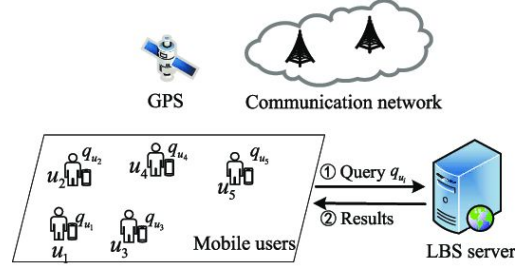


Fig. 1. Users calculating their location and sending it to a LBS

Finally, we assume that the LBS is potentially malicious[3]. This choice is deliberate and reflects a realistic threat model: compromising satellite infrastructure or large-scale communication networks is technically complex and costly, whereas LBS platforms are numerous, heterogeneous, and often managed by third parties. As a result, they represent a more likely attack surface and a statistically more plausible target for data misuse, unauthorized access, or privacy violations. Focusing on a malicious LBS therefore enables a more practical and relevant evaluation of location privacy risks. However, we assume that the server is "Honest-But-Curious" : operates correctly and returns accurate results. While it does not alter the service it provides, it is considered curious or adversarial in the sense that it attempts to infer as much sensitive information as possible from the data it receives.

Expected Results

Beyond quantitative privacy and usefulness metrics, a key expected outcome of this work is improved reproducibility and transparency of experiments on approximate location technique. By providing a publicly and Open-Source accessible web application, the study enables other researchers and users to easily recreate the experiments under identical or modified conditions. Approximate locations can be regenerated directly in the browser, without requiring specialized software or datasets, which lowers the barrier to experimentation.

The web-based approach is expected to facilitate systematic comparisons across different environments, radii, and numbers of approximate points. Users can reproduce figures, validate results, and explore alternative parameter choices in real time. This interactivity also makes it possible to observe how small changes in approximation strategies affect privacy and usefulness.

Overall, the expected result is not only a set of empirical observations, but also a reusable experimental framework that supports replication, extension, and independent verification of results related to location privacy.

II. RELATED WORK

Privacy location isn't a new subject of worry : in fact, since at least 2010, some research papers focusing on it have been released on the internet. In hindsight of the number of mentions of some of these articles, there seems to be major papers that changed the vision of GNSS. While many of them talk about techniques like MAC-Rotation or peer-to-peer anonymizers for location, not much research have been done regarding approximate location and affiliate techniques (like ϵ -geo-indistinguishability).

Still, it is starting to get more recognition as of lately : a significant body of research has been available online since the 2020's, though as it is recent only a few papers are out on the internet, with a lot about the mathematics behind this concept [4]. While user-focused techniques, such as MAC rotation, have received relatively big attention in the literature, there is a much more narrow discussion around the location privacy of new devices, including this new protocol (approximate location) designed to safeguard user data. Few papers create a link between privacy and usefulness of the service. Heitmann et al. [5] conducted a study on approximate-location versus precise location when using LBS on phones (both android and iPhone): they used around 30 applications found both on App store and PlayStore with fine-location then with approximate location and analyzed the changes. From dark pattern (i.e. changing interface to make an user do something) to allowing precise location to bugs in the application, they found out what was the difference between the real feel of use of the application and the new use with approximate location, hence how it affected the user experience. Nevertheless, they have never done a comparison with privacy. This paper [6] is the first to define utility with a set of metrics, even if it focus on LBS using continuous location (RioBuses, Maps-alike). It helps in understanding how it is possible to create measures with few different Location Privacy Preventing Mechanisms (LPPMs) like cloaking, dummy location etc. It looks into some ideas about how a service can run efficiently when the new location is really private.

III. POSSIBLE ATTACKS

When analyzing potential attacks [1] involving unauthorized access to location data, it is essential to consider the granularity of the information. Certain attacks are only feasible when the data meets specific precision requirements. Additionally, the nature of the attack often depends on how the data was accessed. This section will explore both the methods by which location data can be obtained and the types of attacks that may result from such access.

1) *Town-level granularity*: The least precise type of attack involves town or country - level granularity, where the attacker can only determine the receiver's location within hundreds of kilometers. Even at this low resolution, an attacker-using, for example, a malicious weather application - could infer that a family is away on holiday and exploit this information to commit crimes such as burglary or car theft.

2) *District-level granularity - around 10kms*: When an attacker can pinpoint a victim's location within tens of kilometers, they gain district-level accuracy. This information may be obtained through location-based chat application such as dating applications or targeted advertising-like promotions for museums, restaurants, or other local attractions. With such data, an attacker could track the victim's movements, start a surveillance operation, or even sell the location data to third parties. Moreover, this level of detail enables the attacker to analyze the victim's habits, social connections, and relationships, potentially facilitating scams or other malicious activities.

3) *Block-level granularity - around 100m*: Imagine an attacker able to impersonate another user so he could freely ride the automatic toll highways, as the bill would be sent to another user. Another example, an eavesdropper who finds out private information based on location patterns could also resort to blackmailing and produce economical losses and/or social reputation losses. Knowing someone's location with building-level granularity could also lead to public disclosure of unwanted information, such as how often an employee is taking coffee breaks, how often an husband is visiting other places than those he informed his wife about, or the religious and political affiliations of a person (based on the gatherings and places they are attending), etc. In extreme cases, an attacker able to access the location information of many mobile devices simultaneously could easily create a 'heat map' of the most crowded regions in a city at any point of the day and enable various attacks upon crowded areas.

4) *Room-level granularity around 10m*: Many services make use of location data with precision down to a few meters, ranging from home automation systems to applications that offer vouchers when walking by a place or navigating from point A to point B using tools like Google Maps. With access to such detailed information, an attacker could impersonate the victim and potentially carry out any action the victim is authorized to perform.

5) *Professional-level granularity - down to cm*: With the growing adoption of V2X systems, an attacker capable of manipulating data with centimeter-level precision could sabotage these systems entirely. For instance, they might send fraudulent commands to force a lane change and provoke an accident. Alternatively, they could target a specific vehicle (such as a military convoy) by subtly altering its route without the occupants noticing. Furthermore, an attacker could manipulate urban traffic systems to facilitate criminal activities, such as evading law enforcement after committing a crime.

It is clear why location data must remain private and why adjusting granularity for each specific use case is crucial. Even with access to lower-granularity data, malicious actors can still carry out crimes typically associated with higher precision. Therefore, approximate location can be a safeguard for user privacy because even if it is not possible to prevent city-attacks, it is now possible to prevent attacks that require a precision of less than 100m.

IV. USER-SIDE SOLUTIONS

A. Dummy location

Dummy location is a privacy-preserving technique that shields users of location-based services by replacing or distorting their real GPS coordinates with false ones in real-time.[7] Rather than exposing a user's exact position, this method generates plausible but inaccurate location data-such as reporting their presence within a broader area or near a landmark-thereby preventing third parties, advertisers, or malicious actors from tracking precise movements or routines. While this approach may slightly reduce the accuracy of services like navigation or local recommendations, it significantly enhances anonymity and complicates the creation of detailed movement profiles. An advanced variation involves generating multiple fake API calls with slightly altered origins and destination points, making it nearly impossible to determine a user's true location or intended route, though this does increase processing costs for service providers.

B. Cloaking

Cloaking is a privacy-enhancing technique that shields users' real-time location by replacing their exact coordinates with generalized or randomized data, ensuring their movements remain confidential. Rather than revealing precise whereabouts, cloaking dynamically

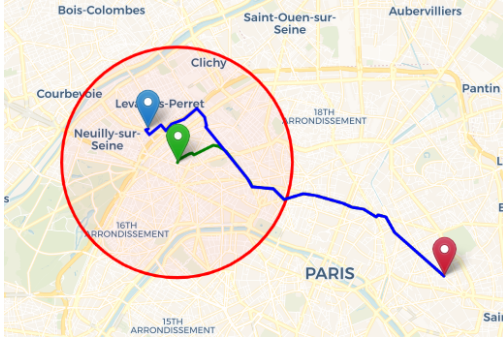


Fig. 2. Real (green) position going to red using approximate location (blue)

obscures a user's position - either by shifting it to a nearby area or blending it with a group of other users[8]. A common approach within cloaking is k-neighbor anonymity, which guarantees that a user's location is indistinguishable from at least k-1 others in the same region, making it statistically impossible to single them out. This method disrupts persistent tracking and prevents the creation of detailed movement profiles, while still allowing location-based services to function, albeit with slightly reduced precision. By combining coordinate distortion with anonymity sets cloaking may introduce minor inaccuracies in location-dependent services, such as navigation or localized alerts. The trade-off ensures robust anonymity and security.

C. Approximate Location

Approximate location is a privacy feature that obscures a user's exact GPS coordinates by rounding their position to a broader area. Both iOS and Android achieve this by dividing the map into grid cells or zones - typically a few kilometers wide - and sharing the coordinates of the center or a fixed point within that cell, ensuring applications receive a consistent but imprecise location. The size of the area can dynamically adjust based on context, such as being smaller in cities and larger in rural regions, to balance privacy and functionality. iOS, since iOS 14 (2020), and Android, since Android 12 (2021)[9], use this method to prevent applications from tracking exact movements or pinpointing sensitive locations. Another idea is to select a random point in a circle where the center is the real position and with a radius that change based on the context of the application (as in figure 2).

V. "APPROXLOC" VISUALIZE PRIVACY VS USEFULNESS

A. Formulas of privacy and usefulness

The privacy and usefulness formulas used in this study were specifically designed for this work. To the best of my knowledge, no existing literature provides a clear or standardized definition of privacy or usefulness in the context of approximate location-based services. As a result, these metrics were formulated to capture intuitive and measurable aspects of anonymity and route accuracy, while remaining simple enough to be implemented and modified within the web application.

1) *Usefulness of a service*: Our first work was to uncover formulas that would combine efficiency and avoidance of any disruption of the service. We propose the following approach:

Utility % = $\frac{\text{Total distance of shared path}}{\text{Total distance of optimal path}} \times 100$ It is also possible to rewrite this formula in regard of the time :

$$\text{Utility}_{time} \% = \frac{\text{Shared path time}}{\text{Optimal path time}} \times 100$$

This formula measures the proportion of the new path that aligns with the theoretically optimal path. The underlying idea is to assess how much the new path deviates from perfection and whether this deviation significantly impacts the user's experience. Indeed the application will give direction and won't be used like google maps, that's why the user is able to know when the path comes close enough to the real location and so can follow the new path. However, the absolute distance to the destination must also be considered. For example, if the destination is nearby, a slightly longer path (e.g., adding 1 minute) may be negligible. On the other hand, if the destination is far, even a small deviation from the optimal path could result in hours of additional travel time. Thus, the utility should account for both the shared path proportion and the relative impact of the deviation on the total travel time.

This formula serves as a first approach to estimate an utility factor and is bound to undergo some improvements.

2) *Privacy of a different location*: To assess anonymity when a user shares a new location, we first need a method to quantify how identifiable they are. Using Bayes theorem, we assume that the probability of being at any given point within a defined area follows a uniform distribution [4]. Consequently, the closer a reported location is to the user's true location, the lower their anonymity. To formalize this, we propose the following privacy metric:

$$\text{Privacy \%} = 1 - \exp(-r \times d(A, C))$$

In this formula, the radius is the one used in the approximate location method, while $d(A, C)$ represents the Haversine distance [10] between the true location A and the reported location C and r the radius used in the approximate location. The server, aware of the radius r, can infer that a larger radius enhances privacy, as it broadens the possible area where the user's true location might lie. Similarly, a greater distance $d(A, C)$ between the true and reported locations further increases privacy, as it reduces the likelihood of accurately identifying the user's position. An interesting scenario arises when the radius is large, but the reported location C is close to the true location A. Even in this case, the server recognizes that the wide radius ensures high privacy, as the user's true location could still be anywhere within the large area defined by the radius.

B. Creation of the WebApp

The real interest of this paper is to present <https://r4ph3uz.github.io/EP2510/> , a WebApp that facilitates the creation of figures and ensures that the experiments are easily reproducible. I developed this web-based application to allow anyone to perform the same analyses and explore additional insights. The application was built using React, Leaflet, and ShadCN, and relies on API calls to OpenStreetMap and Project OSRM for mapping and routing functionalities.

The interface is organized into four main components: the OpenStreetMap layer serving as the background map, an address input panel located in the top-right corner, a settings panel in the bottom-left corner, and the visual elements displayed on the map, including markers and routes. Users can select a specific point on the map to represent their real location, as well as a destination they wish to reach. Once these are defined, a blue marker is generated to represent the approximate location.

Users can adjust the number of approximate-location points and the radius of approximation, and immediately observe the resulting changes in the displayed privacy percentage and usefulness percentage. This interactive setup makes it possible to systematically study the trade-off between privacy protection and practical utility.

C. Methods to get results

Several factors must be taken into account. First, the location of the points under study is crucial, since the impact of using an approximate location varies significantly depending on whether the real position is in a dense urban area or in a rural environment. Privacy risks and practical usefulness are not affected in the same way in a city as they are in the countryside. For this reason, the analysis will focus on three distinct contexts: a large city (such as Stockholm or Paris), a suburban area (such as Täby or Meudon), and a rural setting.

For each of these three locations, a radius of approximation will be selected and systematically adjusted in order to identify an optimal balance between privacy protection and data usefulness.

Finally, to ensure that the results are not overly specific or dependent on isolated cases, the experiment will be repeated multiple times using a large number of points, allowing meaningful averages to be computed.

D. Results

1) *Experiment in cities:* In a dense urban environment, the radius of approximation does not need to be very large to achieve a satisfactory level of anonymity, as a high number of individuals are located within a small geographical area. This natural density significantly increases the anonymity set for a given approximate location.

For the experiment conducted in Paris, the real location was set at *Rond-Point de l'Étoile* (48.8742, 2.2940) and the destination at *Notre-Dame* (48.8534, 2.3487), two well-known landmarks located within the city. The distance between these two points is approximately 5,157 meters. These locations were deliberately chosen because they are representative of a dense urban setting and ensure that the experiment reflects realistic intra-city travel conditions.

TABLE I
ROND-POINT DE L'ÉTOILE - NOTRE-DAME (5KM)

Radius	Usefulness Paris	Privacy Paris
500	93.54%	8.86%
1000	85.68%	35.46%
3000	86.11%	88.33%
5000	61.00%	93.44%
10000	53.07%	100.00%

For the Stockholm experiment, the selected locations were *KTH Entré* (59.3460, 18.0726) and *Monteliusvägen* (59.3203, 18.0625). Both are well-known points of interest within the city and are sufficiently far apart to allow meaningful comparisons with other urban experiments.

TABLE II
KTH ENTRÉ - MONTELIUSVÄGEN (6KM)

Radius	Usefulness KTH	Privacy KTH
500	85.45%	7.43%
1000	71.92%	34.27%
3000	54.59%	88.98%
5000	53.30%	93.70%
10000	30.75%	100.00%

The privacy metric follows a very specific and predictable curve. This is because privacy is determined solely by the number of points contained within a circular area defined by a given radius. As the radius increases, the area of the circle grows, which naturally includes points that can go to the radius and therefore increases the anonymity

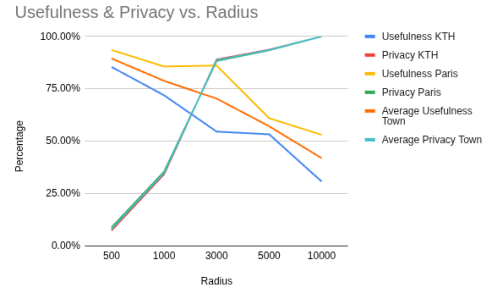


Fig. 3. Usefulness and Privacy over Radius of Approximate Location in city

set. Consequently, privacy becomes essentially a direct function of the radius.

As a result, this behavior is largely independent of the specific experiment or geographic context. While absolute values may vary slightly depending on location density, the overall shape of the privacy curve remains consistent across different graphs.

2) *Experiment in suburbs:* To study suburban environments, locations situated in different neighboring towns were selected, as population density and spatial distribution differ significantly from dense city centers. In France, the experiment uses *Château de Versailles* (48.8032, 2.1264) as the real location and the *Dassault Systèmes headquarters in Vélizy* (48.7860, 2.2107) as the destination. These two points are farther apart than in the urban experiments and lie in separate municipalities, which makes it possible to evaluate the impact of larger approximation radii and to observe how privacy and usefulness evolve in a less dense setting. In Sweden, the experiment was conducted using *Uppsala* (59.8653, 17.6325) as the real location and *Sigtuna* (59.6172, 17.7243) as the destination. These two towns are approximately 35 km apart. Despite this relatively large distance, neither the usefulness nor the privacy metrics changed significantly compared to the experiment near Paris.

Notably, an approximation radius of 3 km proved to be particularly effective, yielding approximately 96% privacy and 82% usefulness. Given these unexpectedly strong results, a potential statistical artifact due to randomness was considered. To address this concern, the experiment was repeated five times, and in all cases both privacy and usefulness remained above 60%, confirming the robustness of the observed outcome.

TABLE III
VERSAILLES - VELIZY (9KM)

Radius	Usefulness France	privacy France
500	93.54%	8.86%
1000	75.31%	35.58%
3000	50.00%	95.04%
5000	55.30%	91.11%
10000	36.67%	100.00%

TABLE IV
UPPSALA - SIGTUNA (35KM)

Radius	Usefulness Sweden	privacy Sweden
500	97.83%	14.07%
1000	85.03%	34.13%
3000	81.55%	96.12%
5000	39.66%	99.60%
10000	26.38%	100.00%

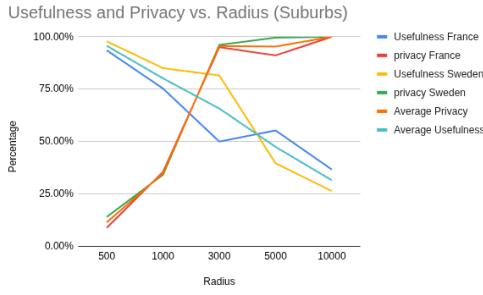


Fig. 4. Usefulness and Privacy over Radius of Approximate Location in suburbs

The difference between city and suburban environments is relatively small in this study, mainly because both contexts still correspond to urbanized areas in terms of the required approximation radius. The selected towns have spatial extents larger than 10 km, which is already the maximum radius considered in this analysis. In suburban settings, users more frequently need to move between different towns, which typically involves using larger roads or highways instead of dense urban street networks. Nevertheless, even in these cases, multiple alternative routes usually remain available, allowing the user to reach the destination through different paths, similarly to what is observed in city-center experiments.

3) *Experiment in countryside:* In rural environments, the experiments reveal a markedly different behavior compared to urban and suburban settings. Due to the low population density and the scarcity of alternative routes, achieving a high level of usefulness is almost always possible. Whether the approximation radius is small or large, the computed routes tend to remain very similar, as they typically rely on the same major roads that minimize travel time. As a result, usefulness remains consistently high across different radii.

Nevertheless, this stability in usefulness comes at the expense of privacy. Because there are fewer plausible starting locations and limited route diversity, increasing the radius does not significantly expand the anonymity set. Consequently, a malicious LBS can more easily infer the user's real location. This highlights an important asymmetry in rural settings: while route accuracy is largely preserved, privacy protection is substantially weaker than in denser environments.

TABLE V
RETHEL - LAON (65KM)

Radius	Usefulness France	privacy France
500	93.67%	6.35%
1000	89.82%	35.87%
3000	78.73%	92.54%
5000	78.23%	91.11%
10000	68.36%	100.00%

TABLE VI
GRÄNGESBERG - KARLSSTADT (155KM)

Radius	Usefulness Sweden	privacy Sweden
500	99.92%	14.13%
1000	99.86%	23.15%
3000	99.38%	87.05%
5000	99.24%	99.94%
10000	85.35%	100.00%

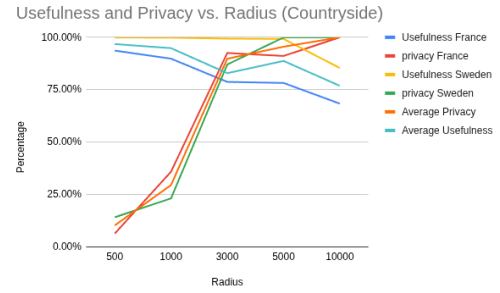


Fig. 5. Usefulness and Privacy over Radius of Approximate Location in countryside

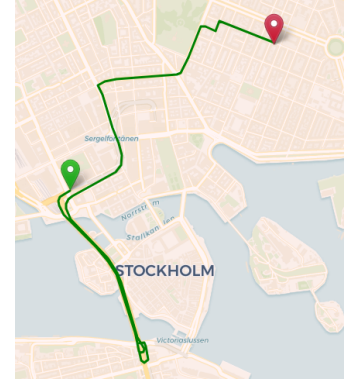


Fig. 6. Take a detour

4) *Specific results:* When points are generated randomly, unfavorable configurations may occasionally occur, which can affect the results. To mitigate this variability, all experiments are therefore conducted using ten points, ensuring more stable and representative outcomes. In addition, in coastal areas such as Stockholm, randomly generated points may fall over water. In such cases, similarly to points located in fields or other non-navigable areas, the routing algorithm computes the shortest path starting from the nearest reachable road segment. This behavior is taken into account in the analysis and does not invalidate the results.

Sometimes it seems that blue or green points don't take the shortest path 6, however it can be explained by the fact that they are on a one-way road, so in order to turn around it makes gigantic detour using round-about, U-turn, exits, etc. This means it is not due to a default in the shortest path algorithm or my webapp, but to the location of the point.

With randomly generated points, certain specific configurations can occasionally be observed. One such case occurs when a distant approximate location (blue point) lies on a route that passes through the real location (green point). In this situation, usefulness reaches its maximum, since the real location lies directly on the computed path and can therefore fully benefit from the routing information. At the same time, privacy is also maximized because both the approximation radius and the distance between the real and approximate locations are large (figure 7).

Another noteworthy configuration arises when the approximation radius is large but the blue point happens to be located close to the real location. In this case, usefulness remains high because the route taken from the approximate location closely matches the real route. Privacy also remains high, not because of spatial separation, but due

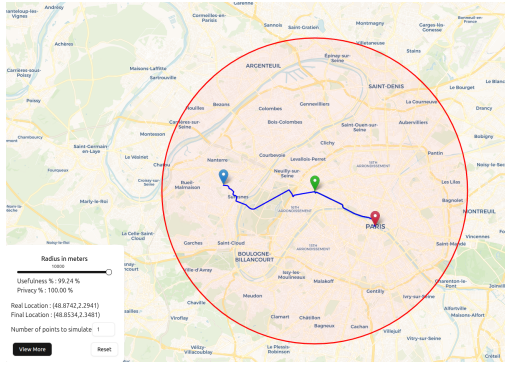


Fig. 7. Approximate Location path going through real location, 100% privacy and usefulness

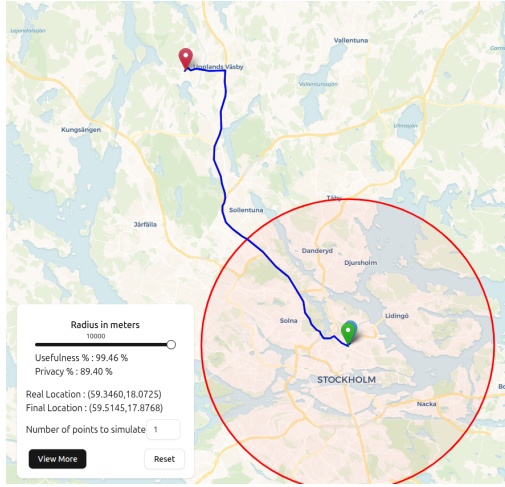


Fig. 8. Approximate Location close to real location with wide radius, 90% privacy and 100% usefulness

to the wide radius, which still provides a large anonymity set despite the proximity of the two points (figure 8).

VI. CONCLUSION

This work investigated the trade-off between privacy and usefulness when using approximate locations in location-based services, with a particular focus on possible threats and how approximate location varies across urban, suburban, and rural environments. The experiments demonstrate that population density and route diversity play a key role: small approximation radii are sufficient in cities, suburban areas behave similarly to urban settings, and rural environments preserve usefulness but offer weaker privacy guarantees. A major strength of this study lies in the open-source web application developed to support it. The tool enables full reproducibility of experiments and allows users to modify parameters and formulas, making it easy to adapt the analysis or test alternative privacy and usefulness metrics. This flexibility encourages further experimentation and independent validation. However, this study also has limitations. Experiments were conducted only in France and Sweden, and results may differ in countries with other urban structures or transportation networks. In addition, approximate locations were generated randomly within a circular area; alternative approaches, such as snapping points to a spatial grid instead of random placement, could yield more stable or realistic results. As a possible extension,

future work could explore adaptive approximation mechanisms that adjust both radius and placement strategy dynamically based on the surrounding environment and user mobility patterns.

REFERENCES

- [1] G. W. Hein, "Location privacy: Challenges and solutions," *Inside GNSS*, 2023, september 19, 2017. [Online]. Available: <https://insidegnss.com/location-privacy-challenges-and-solutions/>
- [2] C. Wang, P. Zhao, H. Huang, R. Zhang, and W. Zhu, *Privacy-Preserving Techniques for the 5G-Enabled Location-Based Services*, 05 2019, pp. 269–299.
- [3] H. Jiang, J. Li, P. Zhao, F. Zeng, Z. Xiao, and A. Iyengar, "Location privacy-preserving mechanisms in location-based services: A comprehensive survey," vol. 54, no. 1, 2021. [Online]. Available: <https://doi.org/10.1145/3423165>
- [4] M. E. Andrés, N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, "Geo-indistinguishability: differential privacy for location-based systems," in *Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security - CCS '13*, ser. CCS '13. ACM Press, 2013, p. 901–914. [Online]. Available: <http://dx.doi.org/10.1145/2508859.2516735>
- [5] S. Heitmann, A. Pagotto, and C. Kray, "Approximate vs. precise location in popular location-based services," *Journal of Location Based Services*, vol. 19, no. 1, pp. 1–42, 2025. [Online]. Available: <https://doi.org/10.1080/17489725.2024.2310006>
- [6] L. Fan and I. Gote, "A closer look: Evaluating location privacy empirically," in *Proceedings of the 29th International Conference on Advances in Geographic Information Systems*, ser. SIGSPATIAL '21. New York, NY, USA: Association for Computing Machinery, 2021, p. 488–499. [Online]. Available: <https://doi.org/10.1145/3474717.3484219>
- [7] S. Zhang, M. Li, W. Liang, V. K. A. Sandor, and X. Li, "A survey of dummy-based location privacy protection techniques for location-based services," *Sensors*, vol. 22, no. 16, 2022. [Online]. Available: <https://www.mdpi.com/1424-8220/22/16/6141>
- [8] H.-J. Lee, S.-T. Hong, M. Yoon, J.-H. Um, and J.-W. Chang, "A new cloaking algorithm using hilbert curves for privacy protection," ser. SPRINGL '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 42–46. [Online]. Available: <https://doi.org/10.1145/1868470.1868480>
- [9] Android. Approximate location. [Online]. Available: <https://developer.android.com/codelabs/approximate-location>
- [10] G. Van Brummelen, *Heavenly Mathematics: The Forgotten Art of Spherical Trigonometry*. Princeton University Press, 2013. [Online]. Available: <https://books.google.fr/books?id=0BCCz8Sx5wkC>